

```
In [1]: import matplotlib.pyplot as plt
```

```
In [2]: import pandas as pd
import seaborn as sns
```

```
In [3]: import numpy as np
```

```
In [4]: data = pd.read_csv("BostonHousing.csv")
```

```
In [5]: data
```

```
Out[5]:
```

	crim	zn	indus	chas	nox	rm	age	dis	rad	tax	ptratio
0	0.00632	18.0	2.31	0	0.538	6.575	65.2	4.0900	1	296	15.3
1	0.02731	0.0	7.07	0	0.469	6.421	78.9	4.9671	2	242	17.8
2	0.02729	0.0	7.07	0	0.469	7.185	61.1	4.9671	2	242	17.8
3	0.03237	0.0	2.18	0	0.458	6.998	45.8	6.0622	3	222	18.7
4	0.06905	0.0	2.18	0	0.458	7.147	54.2	6.0622	3	222	18.7
...	...	...	...	...	...	...	...	...	...	...	...
501	0.06263	0.0	11.93	0	0.573	6.593	69.1	2.4786	1	273	21.0
502	0.04527	0.0	11.93	0	0.573	6.120	76.7	2.2875	1	273	21.0
503	0.06076	0.0	11.93	0	0.573	6.976	91.0	2.1675	1	273	21.0
504	0.10959	0.0	11.93	0	0.573	6.794	89.3	2.3889	1	273	21.0
505	0.04741	0.0	11.93	0	0.573	6.030	80.8	2.5050	1	273	21.0

506 rows × 14 columns



```
In [6]: data.describe()
```

Out[6]:

	crim	zn	indus	chas	nox	rm
count	506.000000	506.000000	506.000000	506.000000	506.000000	506.000000
mean	3.613524	11.363636	11.136779	0.069170	0.554695	6.284634
std	8.601545	23.322453	6.860353	0.253994	0.115878	0.702617
min	0.006320	0.000000	0.460000	0.000000	0.385000	3.561000
25%	0.082045	0.000000	5.190000	0.000000	0.449000	5.885500
50%	0.256510	0.000000	9.690000	0.000000	0.538000	6.208500
75%	3.677083	12.500000	18.100000	0.000000	0.624000	6.623500
max	88.976200	100.000000	27.740000	1.000000	0.871000	8.780000

In [7]: `data.info()`

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 506 entries, 0 to 505
Data columns (total 14 columns):
 #   Column      Non-Null Count  Dtype
---  -
 0   crim        506 non-null    float64
 1   zn           506 non-null    float64
 2   indus        506 non-null    float64
 3   chas         506 non-null    int64
 4   nox          506 non-null    float64
 5   rm           506 non-null    float64
 6   age          506 non-null    float64
 7   dis          506 non-null    float64
 8   rad          506 non-null    int64
 9   tax          506 non-null    int64
10  ptratio      506 non-null    float64
11  b            506 non-null    float64
12  lstat        506 non-null    float64
13  medv         506 non-null    float64
dtypes: float64(11), int64(3)
memory usage: 55.5 KB

```

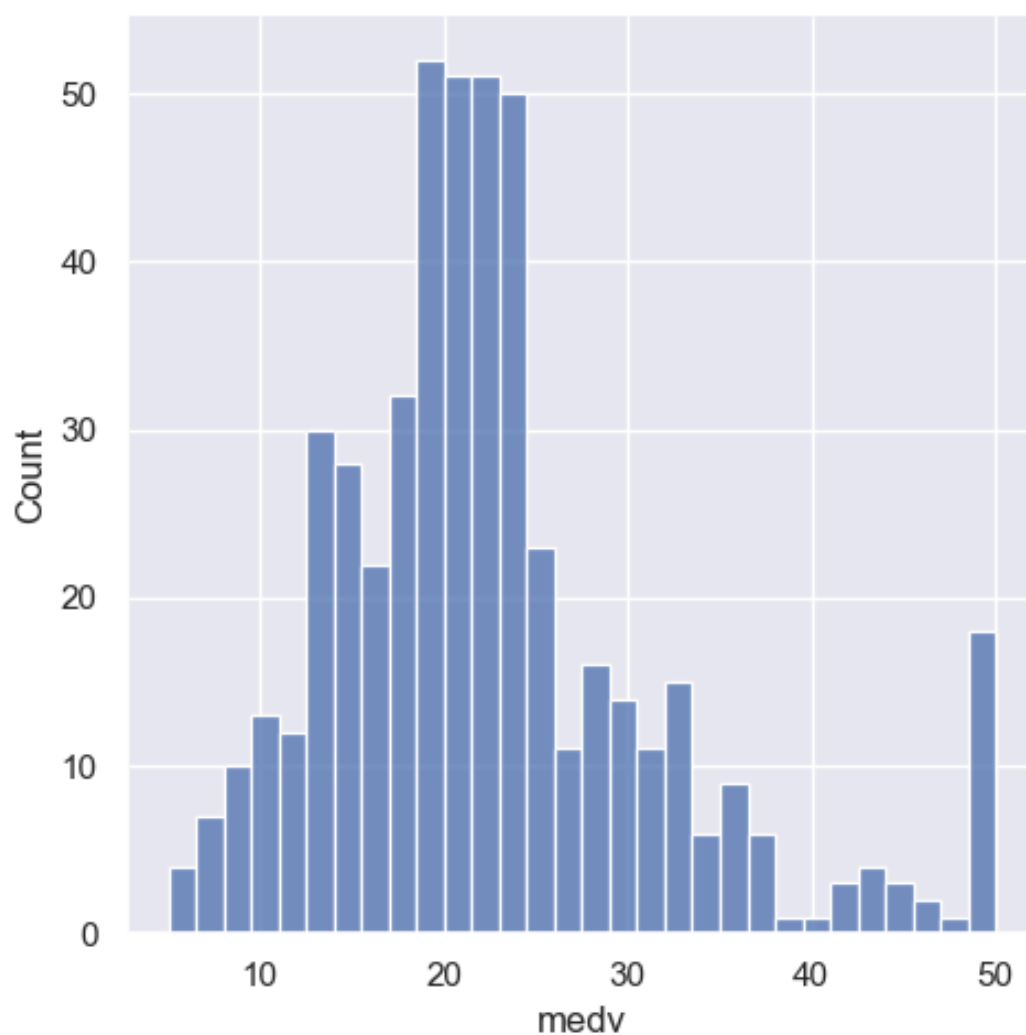
In [8]: `data.isnull().sum()`

```
Out[8]: crim      0
        zn        0
        indus     0
        chas      0
        nox       0
        rm        0
        age       0
        dis       0
        rad       0
        tax       0
        ptratio   0
        b         0
        lstat     0
        medv      0
        dtype: int64
```

```
In [9]: sns.set(rc={'figure.figsize': (11.7, 8.27)})
```

```
In [10]: sns.displot(data['medv'], bins = 30)
```

```
Out[10]: <seaborn.axisgrid.FacetGrid at 0x183e4a74e80>
```



```
In [11]: correlation_matrix = data.corr().round(2)
        sns.heatmap(data=correlation_matrix, annot = True);
```



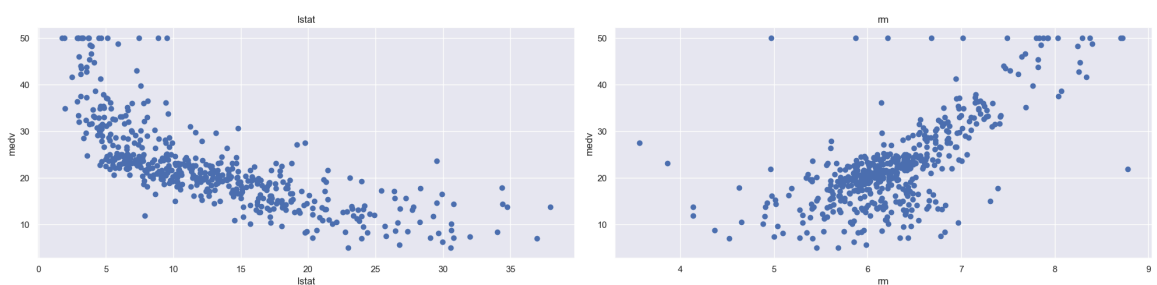
```
In [12]: import matplotlib.pyplot as plt

plt.figure(figsize=(20, 5))

features = ['lstat', 'rm']
target = data['medv']

for i, col in enumerate(features):
    plt.subplot(1, len(features), i + 1)
    x = data[col]
    y = target
    plt.scatter(x, y, marker='o')
    plt.title(col)
    plt.xlabel(col)
    plt.ylabel('medv')

plt.tight_layout()
plt.show()
```



```
In [13]: X = pd.DataFrame(np.c_[data['lstat'], data['rm']], columns=['lstat', 'rm'],  
y = data['medv']
```

```
In [14]: from sklearn.model_selection import train_test_split  
  
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2
```

```
print(X_train.shape)  
print(X_test.shape)  
print(y_train.shape)  
print(y_test.shape)
```

```
(404, 2)
```

```
(102, 2)
```

```
(404,)
```

```
(102,)
```

```
In [15]: from sklearn.linear_model import LinearRegression
```

```
model = LinearRegression()  
model.fit(X_train, y_train)  
  
print(model)
```

```
LinearRegression()
```